Enhancing Black-Box Transparency with SHAP and LIME: A Comparative and Practical Review of Explainable AI in Cybersecurity

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ABSTRACT

As machine learning models are increasingly deployed in cybersecurity systems, their black-box nature complicates trust, accountability, and post-incident investigation. Explainable Artificial Intelligence (XAI) addresses these challenges by providing human-interpretable insights into model behavior without sacrificing predictive performance. Using a standard XAI taxonomy, the paper compares Shapley Additive Explanations (SHAP) and Local Interpretable Model-agonistic Explanations (LIME) in terms of theoretical grounding, stability, interaction awareness, and operational suitability for forensic analysis and intrusion detection. The discussion highlights governance and deployment considerations and outlines future research directions that integrate complementary visualization techniques to strengthen interpretability in cybersecurity workflows. The findings motivate a complementary use of SHAP for global oversight and defensible local attribution, with LIME applied selectively for rapid case-based analysis.

CCS CONCEPTS

• **Computing methodologies** → *Reasoning about belief and knowledge*; **Feature selection**; *Information extraction*; Probabilistic reasoning.

KEYWORDS

Explainable Artificial Intelligence (XAI), Shapley Additive Explanations (SHAP), Local Interpretable Model-agonistic Explanations (LIME), Black Box, Global, Local, Interpretability

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1 INTRODUCTION

1.1 The Black Box Problem

Despite the significant advances AI has brought to cybersecurity, the increasing reliance on machine learning models in critical infrastructure has introduced a major dilemma: AI systems are trusted to make security-critical decisions, yet their internal logic remains opaque to human operators. This phenomenon, commonly referred to as the black box problem, limits transparency in how predictions are formed and complicates trust, accountability, and forensic investigation [2, 5]. In security-critical environments, black-box behavior can obscure bias, data poisoning, adversarial manipulation, and model drift following system updates, thereby increasing operational risk [14].

Although such models often achieve high predictive performance, their opacity creates tension between efficiency and cybersecurity requirements for reliability, auditability, and regulatory compliance. In time-sensitive incident response scenarios, limited explainability can delay root-cause analysis and undermine confidence in automated decision-making systems deployed in critical infrastructure [2].

1.2 The Black Box Solution: XAI

Explainable Artificial Intelligence (XAI) addresses the black box problem by making model behavior more transparent and interpretable to human stakeholders. XAI aims to reduce the trade-off between predictive performance and interpretability while supporting trust, debugging, bias detection, and regulatory accountability [5, 11]. Transparency provides visibility into how a model processes inputs and produces decisions, while interpretability focuses on enabling humans to understand and meaningfully reason about those outputs [9, 14]. In cybersecurity, these properties are essential for validating alerts, supporting forensic analysis, and ensuring responsible deployment of automated decision-making systems.

2 A TAXONOMY OF XAI TECHNIQUES

XAI methods are commonly organized into three categories: (i) whether interpretability is *intrinsic* to the predictive model or provided *post-hoc*, (ii) whether the explanation method is *model-specific* or *model-agnostic*, and (iii) whether explanations are intended to describe behavior *locally* (for an individual prediction) or *globally* (across the model or dataset) [11]. These axes are not mutually exclusive; rather, they provide a practical taxonomy for selecting explanation techniques that match the application's needs (e.g., debugging, compliance reporting, operator trust, or scientific insight).

In fault detection and diagnosis settings, the choice of technique is especially consequential because explanations may be used to justify alarms, guide maintenance actions, and support root-cause reasoning [11]. XAI's three classifications can be visually demonstrated by **Figure 1**.

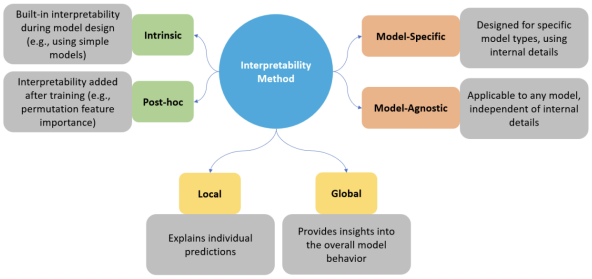

Figure 1: XAI Methods by Categories (Adapted from [11]).

2.1 Intrinsic versus Post-hoc Methods

Intrinsic interpretability refers to models whose structure is directly understandable (e.g., small decision trees, sparse linear models, or rule lists), enabling the model's decision logic to be inspected without an additional explainer [11]. Such models can be advantageous in high-stakes engineering contexts because they reduce the risk that an external explainer will misrepresent model behavior; however, they may underperform when underlying relationships are highly nonlinear or when feature interactions are complex. In contrast, **post-hoc** explainability techniques are applied after a model has been trained, typically treating it as a black box and generating explanations of its outputs [11]. Post-hoc methods, such as LIME and SHAP, are widely used because they can be paired with high-performing models such as gradient-boosted trees or deep neural networks. However, they are additional sources of error and uncertainty, as explanations may be unstable, exhibit limited fidelity to the true decision function, or be sensitive to implementation details such as sampling and feature representation [3, 15].

2.2 Model-Specific versus Model-Agnostic Methods

Model-specific explanation methods exploit internal model information (e.g., gradients, attention maps, or tree structure), which can improve faithfulness for supported architectures but limits generality across model families [11]. In contrast, **model-agnostic** explainers operate solely on inputs and outputs, making them broadly applicable across classifiers and regressors [11]. Both LIME and SHAP are commonly categorized as model-agnostic families in that they can be applied to arbitrary predictive models by querying outputs under controlled perturbations or feature coalition manipulations [11, 16]. Despite this flexibility, model-agnostic methods are often sensitive to design choices: LIME explanations can vary with random seeds, sampling distributions, and locality settings [3, 15], while SHAP explanations raise practical concerns related to computational cost and sensitivity to feature dependence assumptions, particularly when aggregated for global summaries [7, 10].

2.3 Local versus Global Methods

Local explanations justify individual predictions by approximating the model's decision boundary in the neighborhood of an instance, as exemplified by LIME, which fits interpretable surrogates to perturbed samples to explain case-specific outcomes [11]. While this locality supports actionable reasoning in diagnostic workflows, LIME explanations can be unstable across runs or parameterizations, which may undermine operational trust [3, 15]. In contrast, **global** explanations summarize overall model behavior across many instances; SHAP supports both local attributions and global summaries by aggregating Shapley-style feature contributions (e.g., mean absolute SHAP values) to characterize population-level feature importance [10, 16]. This makes SHAP well suited for comparing feature importance, validating model reliance on semantically meaningful signals, and producing model-wide reporting artifacts [11, 12]. Because local and global explanations answer different questions, recent work recommends aligning explainer choice with explanatory scope: using LIME (or stabilized variants) for case-based narratives and SHAP for robust global insight, especially when explanations are aggregated across many predictions [1, 16].

2.4 SHapley Additive exPlanations (SHAP)

SHAP, known formally as **Shapley Additive Explanations**, is an explainability framework grounded in cooperative game theory that attributes a model's prediction to individual input features by computing Shapley values. This formulation provides principled theoretical properties, including local accuracy (completeness) and consistency, which ensure that changes in feature influence are reflected monotonically in attribution scores [6, 13]. A key strength of SHAP is its ability to support both local and global interpretability: at the instance level, SHAP explains individual predictions, while aggregating SHAP values across samples enables global characterization of model behavior, such as identifying dominant predictors and systematic feature effects [6, 11]. This dual scope is particularly valuable in cybersecurity and forensic analysis, where analysts must justify individual alerts while also validating whether model behavior aligns with domain knowledge [11].

Recent applications in intrusion detection and IoT security demonstrate that SHAP supports forensic reasoning by exposing the contribution of traffic features, protocol indicators, and behavioral metrics to classification outcomes [4, 8]. Review studies further emphasize that SHAP enhances trust and transparency in operational pipelines, particularly when explanations must be communicated to analysts and auditors [9, 11]. However, Shapley-based attributions can be computationally expensive for deep learning models and large feature spaces, which may limit scalability in real-time or resource-constrained environments [6, 13]. Consequently, deploying SHAP in production requires balancing its strong theoretical grounding and broad interpretability scope against computational feasibility and operational constraints [9, 11].

2.5 Local Interpretable Model-agnostic Explanations (LIME)

LIME, also known as **Local Interpretable Model-agnostic Explanations**, explains individual predictions by fitting an interpretable surrogate model to perturbed samples generated in the local neighborhood of a target instance. This localized approximation strategy enables case-specific interpretability, making LIME particularly useful for explaining why a given network flow, session, or device state was classified as malicious in cybersecurity and forensic analysis settings [6, 11]. Because LIME treats the predictive model as a black box, it can be applied broadly across diverse architectures, including deep neural networks used in modern intrusion detection systems.

However, LIME does not provide formal theoretical guarantees comparable to those of SHAP, and its explanations depend heavily on the quality of the local surrogate model and the sampling procedure used to generate perturbed instances [13]. The choice of neighborhood definition, kernel weighting, and perturbation strategy can introduce variability across runs and configurations, complicating reproducibility in forensic contexts [4, 6]. Although LIME explanations can be intuitive and actionable for exploratory analysis and rapid incident triage [8], reliance on simple surrogate models may limit its ability to capture complex nonlinear feature interactions, potentially reducing explanatory fidelity in high-dimensional security models [9, 11, 13]. These limitations motivate using LIME cautiously in high-stakes settings and complementing local explanations with more principled methods when explanations are aggregated or audited [6, 11].

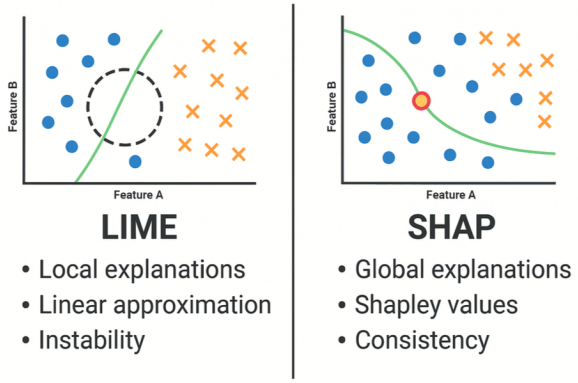

Figure 2: Visual Comparison of SHAP and LIME that highlights core interpretability attributes (Adapted from [6]).

3 COMPARISON

Amid the various approaches of XAI, SHAP and LIME have acquired notability in offering both global and local interpretability [6]. SHAP, with a strong foundation in cooperative game theory, assigns dependable and precise values to the attributes of features. LIME, on the other hand, constructs local surrogate models that approximate the behavior of several intricate classifiers, subsequently promoting the understanding of individual predictions [6]. The interpretability of SHAP and LIME can be visually highlighted in **Figure 2**.

3.1 Using SHAP Globally

Using SHAP for global interpretability involves aggregating feature attributions across a dataset to characterize overall model behavior and identify systematic patterns in feature influence. This global view enables analysts to assess which features consistently contribute to predictions and whether the model's reliance on particular attributes aligns with domain knowledge and operational expectations [6, 11]. Such global summaries are especially valuable in cybersecurity and intrusion detection settings, where understanding population-level drivers of model decisions can support system auditing, policy validation, and model governance workflows [4, 9]. By providing consistent feature attributions grounded in cooperative game theory, SHAP supports principled interpretation of complex models and facilitates communication of model behavior to stakeholders who may not have direct access to model internals [13].

In applied security contexts, global SHAP analyses have been used to examine the relative importance of traffic features, protocol indicators, and behavioral signals in deep learning-based intrusion detection systems, enabling practitioners to validate whether predictions are driven by semantically meaningful features rather than spurious correlations [8]. This population-level perspective also supports comparative evaluation across models and datasets, helping to identify systematic biases or over-reliance on narrow subsets of features that may reduce robustness under distribution shift [6]. Review studies further emphasize that global interpretability can play a critical role in post-incident forensic analysis by contextualizing individual alerts within broader model behavior and trends observed across historical data [9, 11].

Despite these benefits, global SHAP analysis introduces practical considerations related to computational cost and scalability. Aggregating SHAP values across large datasets or high-dimensional feature spaces can be resource-intensive, particularly for deep learning models deployed in real-time or near-real-time security pipelines [7, 13]. Additionally, the interpretation of global SHAP summaries requires careful consideration of feature representation and aggregation strategy, as population-level importance measures may obscure context-specific effects that are salient for individual cases [6, 12]. Consequently, while SHAP provides a robust foundation for global interpretability, its deployment must balance explanatory rigor with operational feasibility in production environments [11].

3.2 Using LIME Locally

Using LIME for local interpretability focuses on explaining individual predictions by approximating the model's behavior in the neighborhood of a specific instance. This localized perspective is particularly valuable in forensic and cybersecurity workflows, where analysts often seek to understand why a particular network flow, session, or device state was flagged as malicious [6, 11]. By fitting simple surrogate models to perturbed samples around a target instance, LIME provides case-specific explanations that can support rapid incident triage, exploratory analysis, and interactive debugging of model outputs [4, 8].

Local explanations generated by LIME are model-agnostic and can be applied across diverse classifier architectures, including

deep learning models commonly used in intrusion detection systems. This flexibility has contributed to the widespread adoption of LIME in applied security research and practice, particularly in settings where transparency is needed for individual alerts without requiring full model retraining or architectural modification [9, 11]. In forensic analysis, LIME explanations can provide intuitive narratives for specific decisions, aiding communication between automated detection systems and human analysts responsible for follow-up investigation [6].

However, the local nature of LIME explanations introduces important limitations. The fidelity of the surrogate model depends on the choice of perturbation strategy, neighborhood definition, and kernel weighting, which can lead to variability across runs and sensitivity to methodological choices [13, 15]. Furthermore, because LIME typically employs simple surrogate models, it may fail to capture complex nonlinear interactions among features that influence predictions in deep learning-based security systems [9, 11]. Reliability studies also highlight that local post-hoc explanations can exhibit instability under repeated sampling, raising concerns about reproducibility and interpretive consistency when explanations are used in high-stakes or legally scrutinized forensic contexts [3, 6]. As a result, while LIME provides valuable instance-level insight, its outputs should be interpreted with caution and are often most effective when complemented by global interpretability methods such as SHAP [1].

3.3 Local Comparison: SHAP versus LIME

Although SHAP is frequently used for global model interpretation, it also supports instance-level explanations through local feature attributions. In cybersecurity and forensic contexts, local SHAP explanations can be used to justify individual alerts in a manner similar to LIME, enabling analysts to inspect the contribution of specific traffic features or behavioral indicators to a single prediction [6, 13]. The theoretical grounding of SHAP provides consistency in local attributions, which can improve confidence in explanation stability when individual decisions are scrutinized in post-incident analysis [11].

At the local level, both SHAP and LIME aim to support case-based reasoning; however, they differ in how explanations are constructed and interpreted. LIME approximates model behavior in the neighborhood of a specific instance using a surrogate model, which can yield intuitive explanations but is sensitive to sampling strategies and locality definitions [3, 15]. SHAP, in contrast, directly attributes contributions to features using a principled allocation scheme, reducing dependence on local surrogate fidelity [13]. Empirical studies in intrusion detection indicate that while both methods can produce meaningful local explanations, SHAP explanations tend to exhibit greater stability across configurations, whereas LIME explanations may vary with methodological choices [6, 8].

From a practical standpoint, SHAP may be preferred for local explanation in high-stakes forensic scenarios where explanation consistency and defensibility are critical, while LIME may remain useful for rapid, exploratory analysis during early-stage incident triage [4, 9]. This distinction highlights the importance of aligning local explanation techniques with operational requirements, particularly when explanations are used to support decisions with legal or regulatory implications [11].

4 DISCUSSION

4.1 Operational Integration of LIME and SHAP in Cybersecurity Workflows

In operational cybersecurity environments, explainability methods should support rapid incident response, analyst sensemaking, and post-incident forensic investigation. Local explanation methods such as LIME are well suited for day-to-day triage workflows, where analysts seek to understand why a particular network flow, session, or device state triggered an alert [6, 8]. By providing case-specific explanations, LIME can assist in validating alerts, prioritizing responses, and guiding immediate investigative actions. However, operational use of LIME should be accompanied by standardized configuration and documentation of perturbation and sampling parameters to reduce variability and improve interpretive consistency [13, 15].

Global explanation methods such as SHAP complement these workflows by enabling analysts to monitor model behavior over time, identify shifts in feature importance, and detect emerging patterns in alert generation [4, 11]. Integrating SHAP into monitoring dashboards or periodic model review processes can support proactive assessment of model drift, feature dependency changes, and systematic vulnerabilities in detection pipelines [9]. Together, local and global explanations support both immediate operational needs and longer-term situational awareness in cybersecurity systems.

4.2 Governance and Auditability in XAI Systems

Explainability plays a central role in governance and compliance frameworks for AI-enabled cybersecurity systems. Global interpretability using SHAP is particularly well suited for audit and oversight processes, as aggregated feature attributions provide transparent documentation of model behavior across datasets and deployment contexts [6, 11]. Such summaries can support internal audits, external review, and regulatory reporting by demonstrating that model decisions are grounded in semantically meaningful and operationally relevant features [4, 9].

Forensic defensibility further requires that explanations be reproducible and methodologically justified. SHAP's principled theoretical foundation supports consistent attribution of feature contributions, which strengthens the evidentiary value of explanations used in post-incident analysis or compliance documentation [13]. In contrast, when LIME is used for governance-related purposes, organizations should establish protocols for explanation generation and validation to mitigate instability and variability across runs [3, 15]. Clear documentation of explanation methodology enhances accountability and reduces the risk of misinterpretation in legally or operationally consequential settings [11].

4.3 Deployment Considerations for Global Interpretability

Global explanation methods are most appropriate when the objective is to understand overall model behavior, validate feature

relevance at the population level, or assess alignment between model logic and domain knowledge. SHAP is particularly well-suited for these purposes due to its ability to aggregate feature attributions across datasets and provide consistent measures of global feature importance [6, 12]. In cybersecurity applications, global SHAP analysis can reveal whether detection models systematically rely on meaningful traffic characteristics or inadvertently emphasize spurious correlations [4, 8].

Global explanations are also valuable for comparative evaluation across models and configurations, enabling practitioners to assess how changes in architecture or training data affect feature reliance and decision logic [9, 11]. Additionally, SHAP-based global summaries support post-deployment monitoring by highlighting shifts in feature importance that may indicate model drift or evolving threat patterns [7, 13]. Accordingly, global explanations should be prioritized in model development, validation, and governance workflows where population-level insight is required.

4.4 Deployment Considerations for Local Interpretability

Local explanation methods are most appropriate when the objective is to justify or interpret individual predictions in operational or forensic contexts. LIME is well suited for explaining specific alerts or classifications, enabling analysts to understand the localized factors that influenced a particular decision [6, 11]. Such instance-level explanations support rapid triage, hypothesis generation, and interactive debugging of detection systems during live incident response [8].

Local explanations are also useful in user-facing contexts where explanations must be communicated to non-technical stakeholders responsible for operational decisions or follow-up investigation [4]. However, given the sensitivity of LIME explanations to sampling and neighborhood definitions, local explanations should be interpreted as approximations rather than definitive representations of model logic [13, 15]. In high-stakes scenarios, local explanations should be corroborated with global interpretability methods or repeated under controlled configurations to improve reliability [3].

4.5 Combining Global and Local XAI Methods

Effective deployment of XAI in cybersecurity systems benefits from combining global and local explanation methods to address complementary interpretability needs. Global SHAP analysis provides population-level insight into model behavior, supporting auditing, governance, and validation, while local LIME explanations support case-based reasoning and operational triage [1, 6]. Integrating both perspectives enables practitioners to contextualize individual alerts within broader patterns of model behavior, strengthening situational awareness and forensic reasoning [9, 11].

From a practical standpoint, organizations should adopt structured workflows in which global interpretability is used for periodic model review and governance, while local explanations are generated on demand during incident response. Such workflows should include standardized explanation protocols, documentation of methodological choices, and validation procedures to ensure reproducibility and interpretive consistency [13, 15]. This complementary deployment strategy aligns each method with its respective strengths and mitigates the limitations of relying on any single explainability technique in isolation [4].

4.6 Interpretability versus Performance

Deploying explainability methods in production cybersecurity systems requires careful management of trade-offs between interpretability, computational cost, and real-time performance. Global SHAP analysis can be computationally intensive when applied to large datasets or deep learning models, which may limit its feasibility for continuous monitoring in high-throughput environments [7, 13]. Organizations should consider batch-based or periodic SHAP analysis for governance and auditing purposes, reserving real-time computation for critical investigative scenarios [11].

Similarly, while LIME is comparatively lightweight for individual explanations, repeated local explanation generation can introduce latency in time-sensitive workflows. Practical deployment should therefore incorporate caching, explanation sampling strategies, or selective explanation triggers to balance interpretability needs with operational performance constraints [9]. Managing these trade-offs ensures that explainability enhances, rather than hinders, the effectiveness and responsiveness of cybersecurity systems in real-world deployment contexts.

4.7 Enhancing Global SHAP with Complementary Visualizations

While SHAP provides robust global interpretability through aggregated feature attributions, global explanation alone may not fully capture how feature effects vary across input ranges or operational contexts. Complementary global visualization techniques, such as PDP (Partial Dependence Plots), may help contextualize SHAP-derived importance measures by illustrating how changes in specific features influence model outputs across their observed distributions. In cybersecurity applications, such visualization could support deeper understanding of how traffic characteristics or behavioral indicators affect detection outcomes under varying operational conditions [9, 11].

Combining SHAP-based global importance with complementary visualization supports more nuanced model validation and interpretation. This layered view helps distinguish features that are broadly influential from those whose effects are context-dependent, reducing the risk of misinterpreting aggregated importance measures. In addition, visual summaries of feature-response relationships can improve communication of model behavior to non-technical stakeholders involved in governance and oversight, strengthening transparency in cybersecurity decision-making processes [9, 11].

4.8 Enhancing Local LIME with Global Interpretability Methods

Local explanations produced by LIME support instance-level interpretability but do not provide context regarding how individual explanations relate to broader model behavior. Complementing local LIME explanations with global interpretability methods, such as PDP, may help mitigate concerns about surrogate instability and

limited interaction awareness [3, 15]. In cybersecurity workflows, pairing local LIME explanations with global summaries of feature importance or model behavior can provide analysts with both case-specific justification and population-level context, supporting more informed forensic reasoning [6, 8].

This two-tier interpretability strategy reduces over-reliance on potentially unstable local explanations by grounding instance-level narratives in broader patterns of model behavior. In forensic contexts, such contextualization can improve analyst confidence in explanation outputs and support more defensible reasoning by demonstrating consistency between individual explanations and population-level trends [6].

4.9 The Future of XAI: SHAP, LIME or both?

Given that SHAP supports both local and global interpretability and offers stronger theoretical guarantees than LIME, an open question is whether SHAP can fully replace LIME in applied cybersecurity contexts. The ability of SHAP to provide consistent local attributions and aggregated global summaries suggests that a unified SHAP-based explainability framework may reduce methodological complexity in deployment pipelines [6, 13]. However, the computational overhead associated with SHAP, particularly for deep learning models, may limit its practicality for real-time, large-scale systems [7].

However, practical constraints complicate the feasibility of fully replacing LIME with SHAP in operational environments. The computational overhead associated with SHAP, particularly for deep learning models, may limit its applicability in real-time or resource-constrained systems [7]. In contrast, LIME's lightweight surrogate modeling approach enables rapid instance-level explanation generation, which remains advantageous during time-sensitive incident response. These trade-offs suggest that SHAP and LIME are likely to remain complementary rather than mutually exclusive in applied cybersecurity pipelines [6, 13].

5 CONCLUSION AND FUTURE WORK

This paper addressed the black box problem in AI-enabled cybersecurity by reviewing Explainable Artificial Intelligence techniques that improve transparency, trust, and accountability in security-critical environments [2, 5]. Using a standard taxonomy of XAI methods, the analysis focused on SHAP and LIME as representative post-hoc, model-agnostic approaches supporting global and local interpretability in forensic and intrusion detection workflows [6, 11]. The comparison showed that SHAP provides consistent global summaries and defensible local attributions, while LIME offers accessible instance-level explanations but may introduce variability due to sensitivity to sampling and locality definitions [3, 13].

Future work should empirically evaluate the integration of complementary visualization techniques with explanation pipelines in cybersecurity, including PDP (Partial Dependence Plots) for global feature effect analysis and ICE (Individual Conditional Expectation) for instance-level response profiling. Although such methods are established in other application domains, their systematic evaluation in intrusion detection and forensic security contexts remains limited. In addition, further study is needed to assess whether SHAP-based local and global explanations, augmented by complementary visualization techniques, can reduce or replace reliance on local surrogate explanations such as LIME in high-stakes operational environments [6, 9, 11]. These directions offer a path toward more unified, stable, and operationally defensible explainability frameworks for AI-enabled cybersecurity systems.

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